

Chapter 1 of Navidi

Agenda

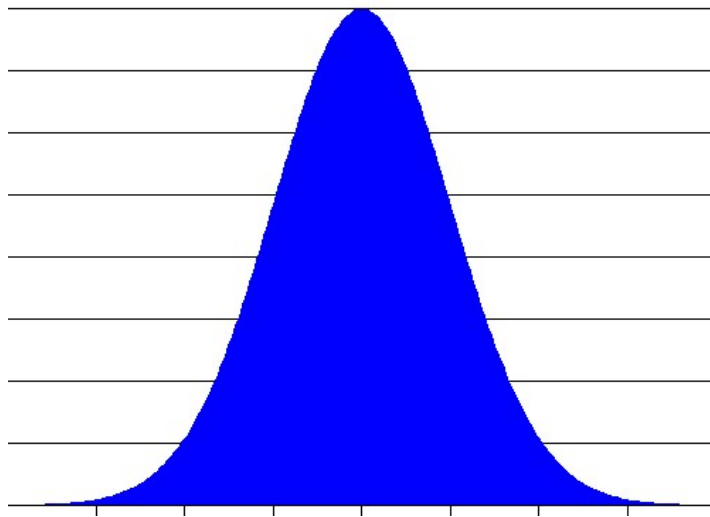
- Population vs samples
- Sampling
- Statistic
- Plots

Why Statistics?

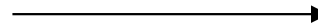
- Deal with uncertainty in repeated scientific measurements.
- Draw reliable conclusions from data.
- Design valid experiments and draw reliable conclusions.
- Be a well-informed member of society.

Population and statistical inferences

A population is the entire group of objects about which you want information.



Population is completely identified: Both the shape and the parameters are known.

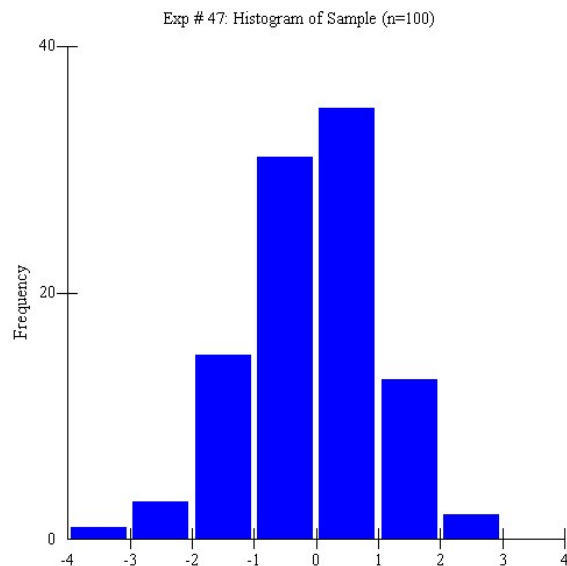


Compute probabilities, expectations, etc., for random variables X from this population.
 X : future observations from this population

Deductive inferences
(Tumdengelim)

Sample and statistical inferences

A sample is a subset of observations selected from the population. It is used to gain knowledge about the population.

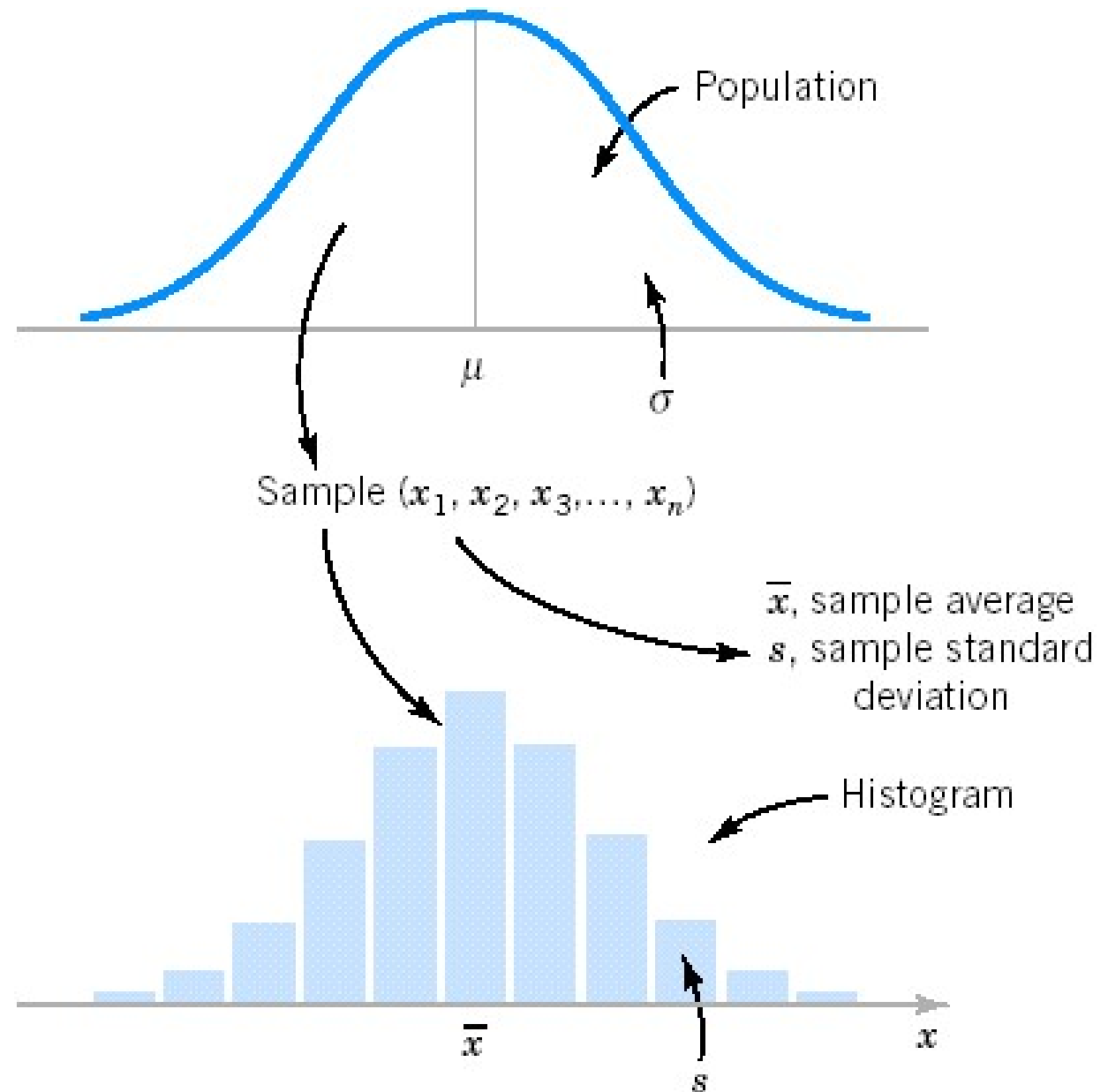


Estimate the unknown parameters of this population.

The shape of the population is known, but not its parameters.

Inductive inferences
(Tumevarim)

Relationship between a population and a sample

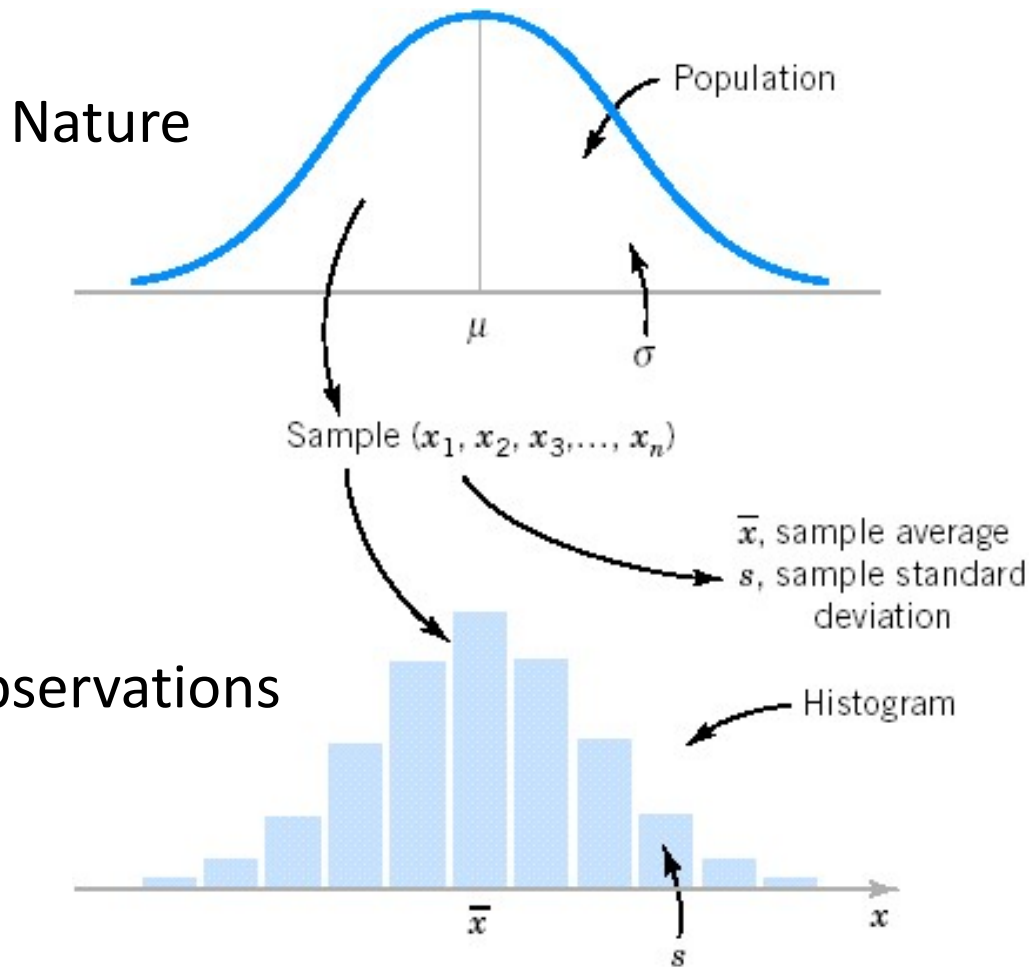


Games online?

- Have you played Tavla online?
- How do you think the dice are rolled?
- Can we also simulate a die roll? → Labs!

How to link the two worlds?

Population and Sample



- If we can play both sides, we can understand the link between them better!
- Simulations where you can generate die rolls, coin tosses and more → representing the God who controls the Nature
- Analyzing the data that you yourself generated will show you how you can use the data to understand the population

In the Labs:

- You will play the God sometimes, and yourselves in other times...
- God: Simulations (many data sets can be generated)
- Yourself: Statistical analysis via Python (just one data set is there to analyze)

In the lectures:

- You will:
 - understand the link between the two worlds mathematically,
 - learn which questions to ask and how to answer them when you have a problem and a corresponding data set.

Example 1 Quality Control

- 1000 rods are produced to meet the specification 0.45 ± 0.02 cm
- No time and no resource to check all 1000 rods.
- What to do?

Here are some relevant questions:

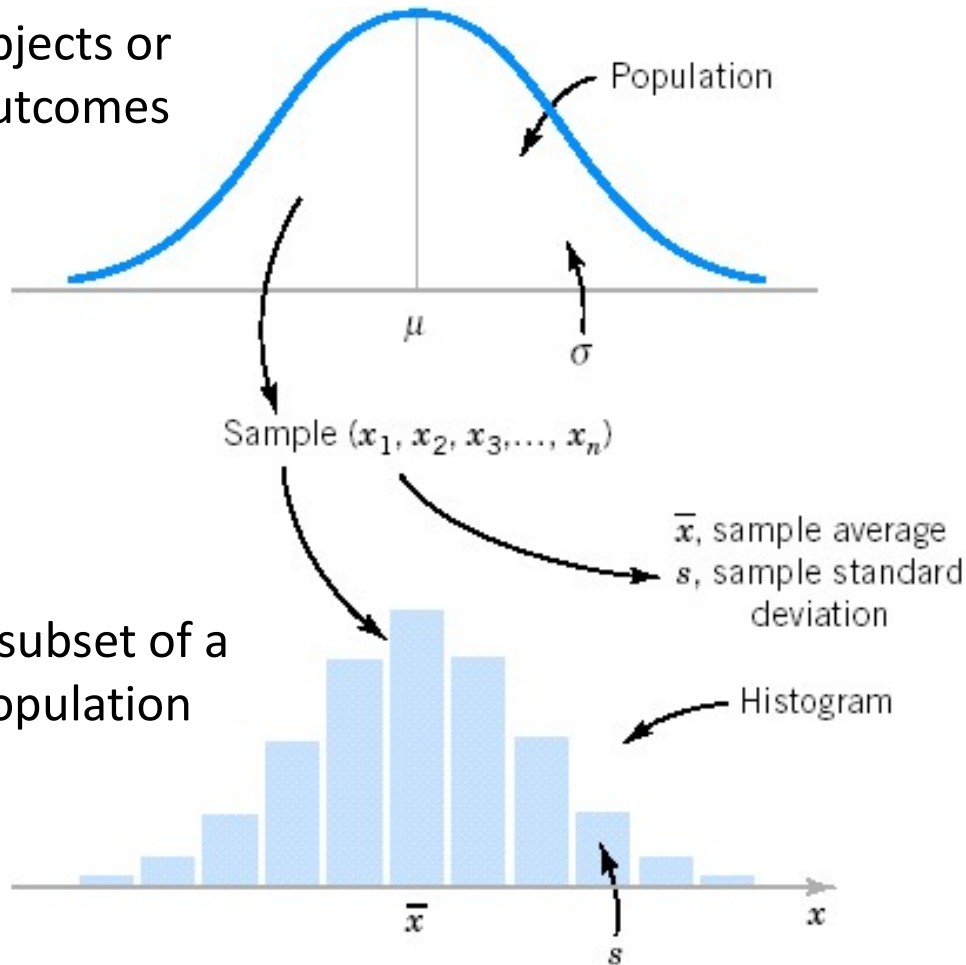
1. How large is a typical difference between the sample proportion and the population proportion for this kind of sample?
2. What interval gives a good estimate of the percentage of acceptable rods in the population with reasonable certainty?
3. How certain can the engineer be that at least 90% of the rods are good?

Statistics can help us to address questions like these.

How to link the two worlds?

Population and Sample

entire collection of
objects or
outcomes



- A **simple random sample (SRS)** of size n : a sample chosen by a method in which each collection of n population items is equally likely to comprise the sample, just as in the lottery.

- The random variables $X_1, \dots, X_n$ are said to **simple random sample** of size n if:
 - (1) X_i 's are independent
 - (2) X_i 's are identical

Independence

- The items in a sample are said to be **independent** if knowing the values of some of the items does not help to predict the values of the others.
→ **choosing one item for the sample has no effect on the choice of other items**
- Items in a simple random sample may be treated as independent in most cases encountered in practice, unless the population is finite, and the sample comprises a substantial fraction (more than 5%) of the population.

How to draw a random sample?

- A professor's question: average height of students at a university?
- Population?
- How to choose a sample of size of 100?

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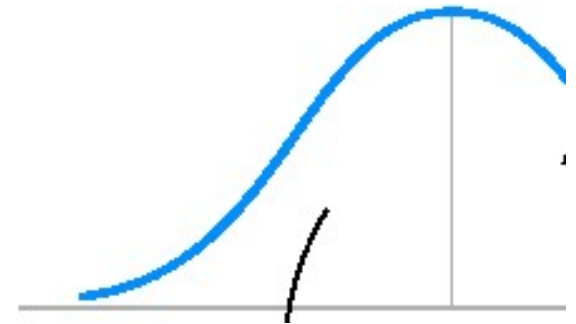
What can be wrong?



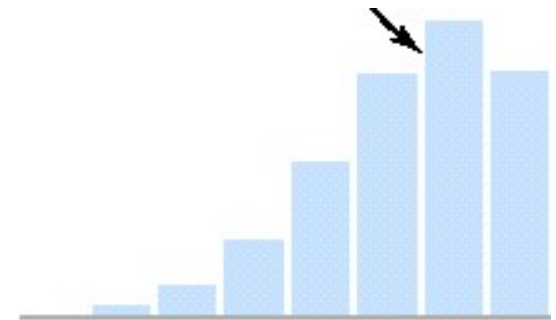
A Random Sample

Definition: The random variables $X_1, \dots, X_n$ are said to form a (simple) random sample of size n if

- (1) X_i 's are independent r.v.s,
- (2) X_i 's have the same probability distribution



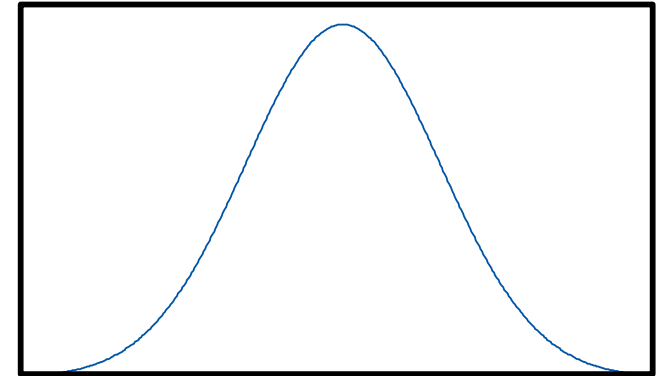
An additional issue is on whether a random sample is representative of the underlying population or not.



A Random Sample

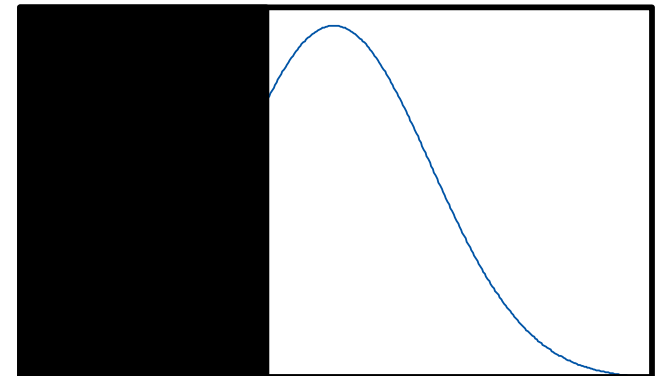
- Should be representative of the whole population.
- How reliable is the data set? Does it represent the whole population?

Population

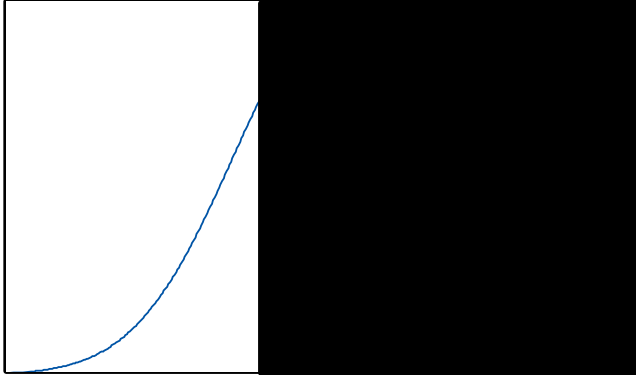


Violates in
independence
assumption.

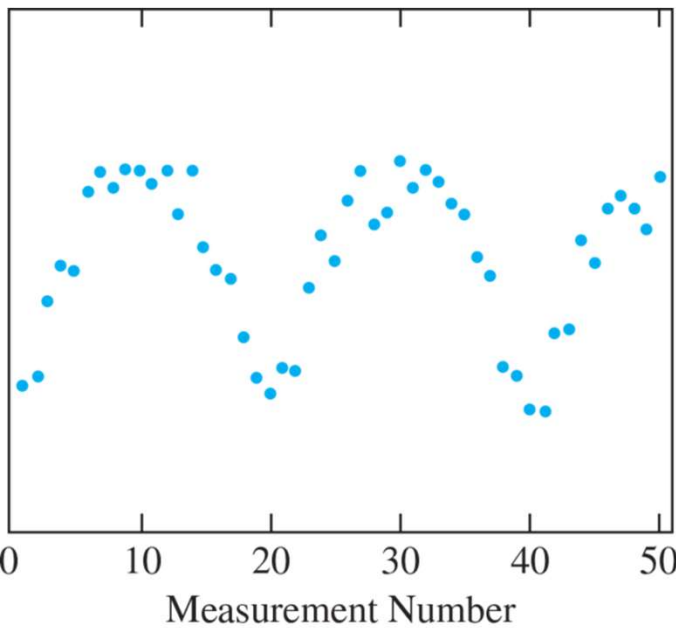
Statistics:
Population
with one in
four poor



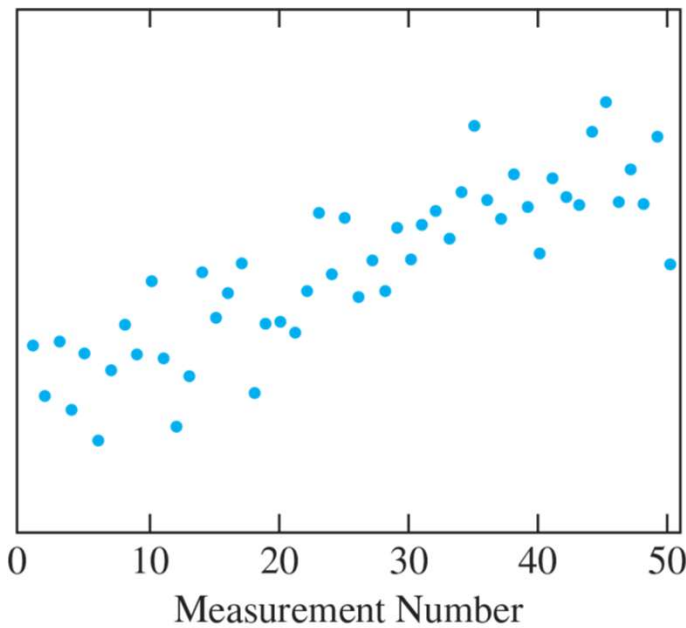
Or:



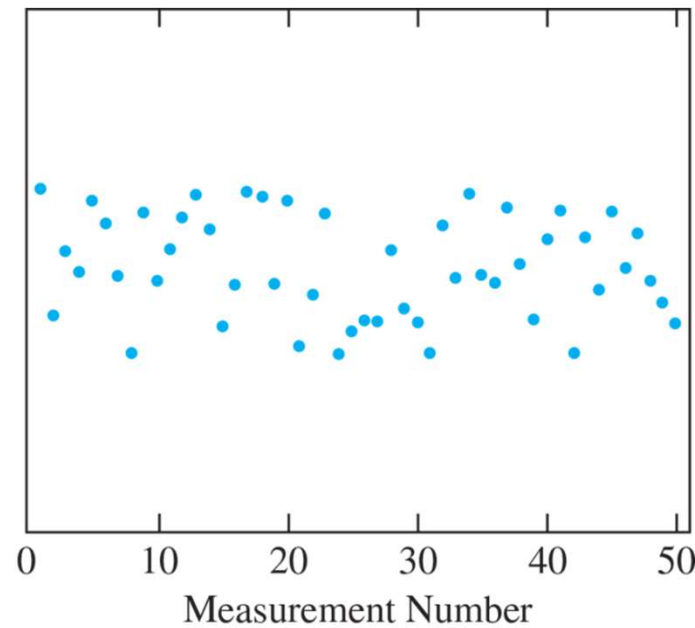
Which One(s) Random Sample?



(a)



(b)



(c)

Simple Random Sampling

A SRS is not guaranteed to reflect the population perfectly.

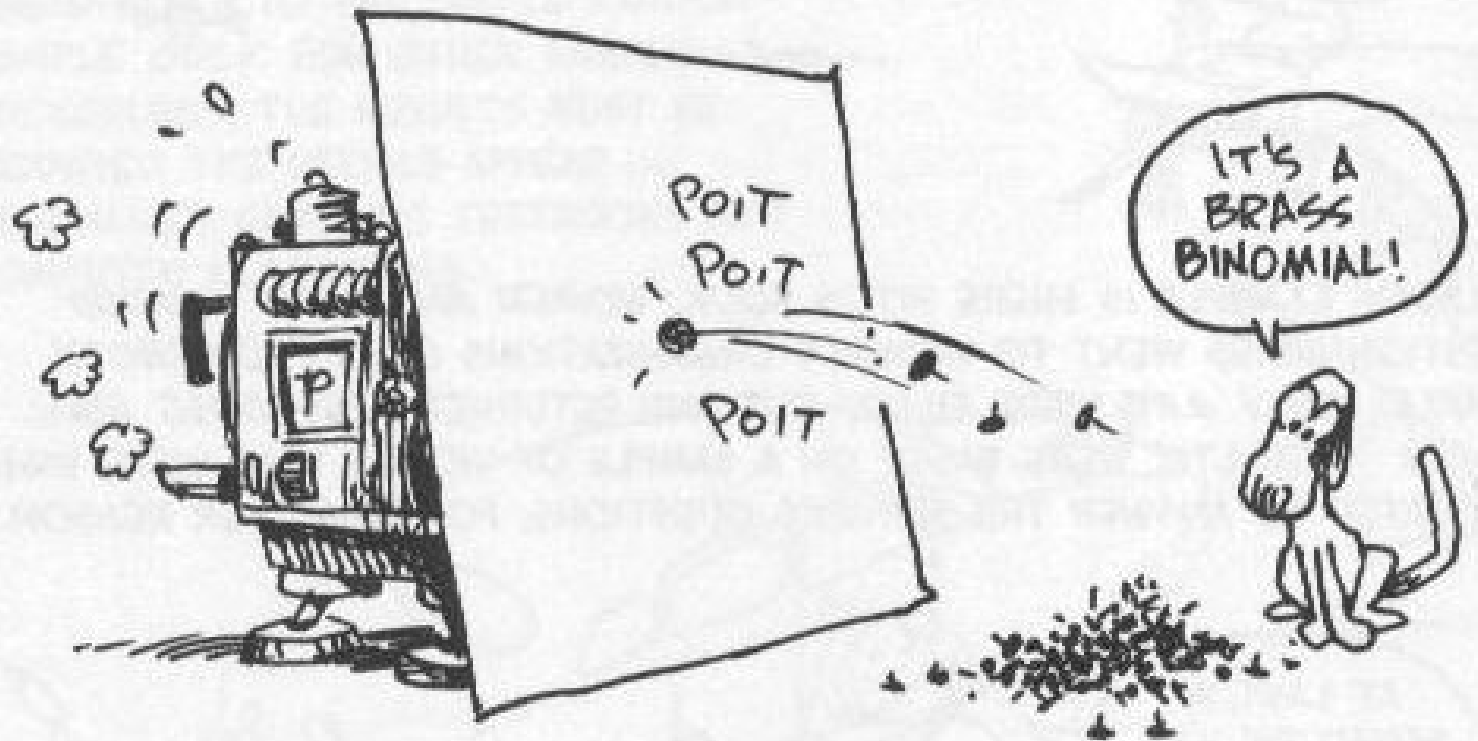
SRS's always differ in some ways from each other; occasionally a sample is substantially different from the population.

Two different samples from the same population will vary from each other as well.

- This phenomenon is known as **sampling variation**.

Tack Machine

THE ASTUTE READER WILL RECOGNIZE THIS AS A *BERNOULLI SYSTEM*: EACH NEW TACK IS THE OUTCOME OF A *BERNOULLI TRIAL* WITH SOME PROBABILITY p OF SUCCESS (I.E., BEING DEFECT-FREE) AND PROBABILITY $1-p$ OF FAILURE (I.E., BEING DEFECTIVE).



WE THINK OF THIS SITUATION AS IF THERE WERE A *HIDDEN BUT REAL "BERNOULLI MACHINE"* WHOSE PROBABILITY p GOVERNS THE OUTCOMES WE OBSERVE IN THE SO-CALLED "REAL WORLD."

Estimate probability of non-defective?

SINCE THE BERNOULLI MACHINE IS INVISIBLE, WE DON'T KNOW WHAT p IS, BUT WE'D LIKE TO FIND OUT. SO WE TAKE A RANDOM SAMPLE OF n TACKS, AND FIND THAT x OF THEM ARE O.K.



HMM... FEELS LIKE $n=400$ AND $x=352$...

NOW THE PROPORTION OF SUCCESSES IN THE SAMPLE SHOULD BE SOMEWHERE AROUND p ... SO WE CALL IT $\hat{p}$, PRONOUNCED "P-HAT."

$$\hat{p} = \frac{x}{n}$$

$\hat{p}$ IS THE NUMBER OF SUCCESSES x IN THE SAMPLE, DIVIDED BY THE SAMPLE SIZE n . FOR EXAMPLE, IF p WAS .85, AND WE SAMPLED $n=1000$ TACKS, MAYBE WE FOUND $x=832$ GOOD ONES, MAKING $\hat{p} = .832$.

If we took many samples of 1000 tacks:



This phenomenon is known as sampling variation.

Example 1 Quality Control

- 1000 rods are produced to meet the specification 0.45 ± 0.02 cm
- No time and resource to check all 1000 rods.
- 50 rods are checked $\rightarrow$ 46 of them met specification $\rightarrow$ 92%
- What can we say about the population of 1000 rods?

Sampling variation $\rightarrow$ the true proportion of rods that meet specification will be close to the sample proportion, or 92%, but not exactly equal.

A conceptual population

All of the values that might possibly have been observed from a population.

- For example, a geologist weighs a rock several times on a sensitive scale. Each time, the scale gives a slightly different reading.
- Here the population is conceptual. It consists of all the readings that the scale could in principle produce.

A sample of convenience

A sample that is obtained in some convenient way and is not drawn by a well-defined random method.

Things to consider:

- Convenience samples may differ **systematically** in some way from the population.
- Thus, they should only be used when it is not feasible to draw a random sample.
- If you do have a convenience sample, think carefully about the ways in which the sample might differ systematically from the population.

How to draw a sample of convenience?

- A professor's question: average height of students at a university?
- Population?
- How to choose a sample of size of 100?

Other Sampling Methods

Samples other than simple random samples can be useful in various situations.

These include (but are not limited to):

- Weighted sampling.
- Stratified random sampling.
- Cluster sampling.

Types of Experiments

Many types of experiments can be used to generate data.

One-sample experiments involve only one population of interest with a single sample drawn from it.

Multi-sample experiments involve two or more populations of interest with a single sample drawn from each population.

- In many multi-sample experiments, the populations can be distinguished from one another by varying one or more **factors** that may affect the outcome of interest.

Types of Data

When a numerical quantity designating how much or how many is assigned to each item in the sample, the resulting set of values is **numerical or quantitative**.

- Height (in centimeters).
- Weight (in kilograms).
- Age (in years).

When sample items are placed into categories and category names are assigned to the sample items, the data are **categorical or qualitative**.

- Hair color.
- Country of origin.
- Zip code.

Controlled Experiments and Observational Studies

A **controlled experiment** is one in which the experimenter controls the values of the factors.

- When designed and conducted properly, controlled experiments can produce reliable information about cause-and-effect relationships between factors and response.

An **observational study** is one where the experimenter simply observes the factors as they are, without having any control over them.

- Observational studies are not nearly as good as controlled experiments for obtaining reliable conclusions regarding cause and effect.

Statistic

- Let $X_1, \dots, X_n$ be a (simple) random sample.
- A statistic is any function of the observations in a random sample.
- A statistic is a random variable, so it has its own probability distribution (generally called sampling distribution).

Examples of statistics

Let $X_1, \dots, X_n$ be a random sample.

- Sample mean
- Sample standard deviation
- Minimum value of the data
- Maximum value of the data
- Mode
- Median
- Quartiles
- ...

How about their distributions?

Measures of central tendency

- Do we mean by “average”?

A shoe manufacturing business

- 3 partners: CEO, production manager, sales manager
- 5 engineers
- 40 workers
- Average salary in this business?

CEO	40000
PM	30000
SM	30000
E1	10000
E2	10000
E3	10000
E4	10000
E5	10000

W1	3000
W2	3000
W3	3000
W4	3000
W5	3000
W6	3000
W7	3000
W8	3000
W9	2500
W10	2500

W11	2500
W12	2500
W13	2500
W14	2500
W15	2500
W16	2500
W17	2500
W18	2500
W19	2500
W20	2500

W21	2000
W22	2000
W23	2000
W24	2000
W25	2000
W26	2000
W27	2000
W28	2000
W29	2000
W30	2000

W31	2000
W32	2000
W33	2000
W34	2000
W35	2000
W36	2000
W37	2000
W38	2000
W39	2000
W40	2000

Definitions:

Measures of central tendencies

- Let $X_1, \dots, X_n$ be a random sample.
- Sample mean is the arithmetic average of all observations.

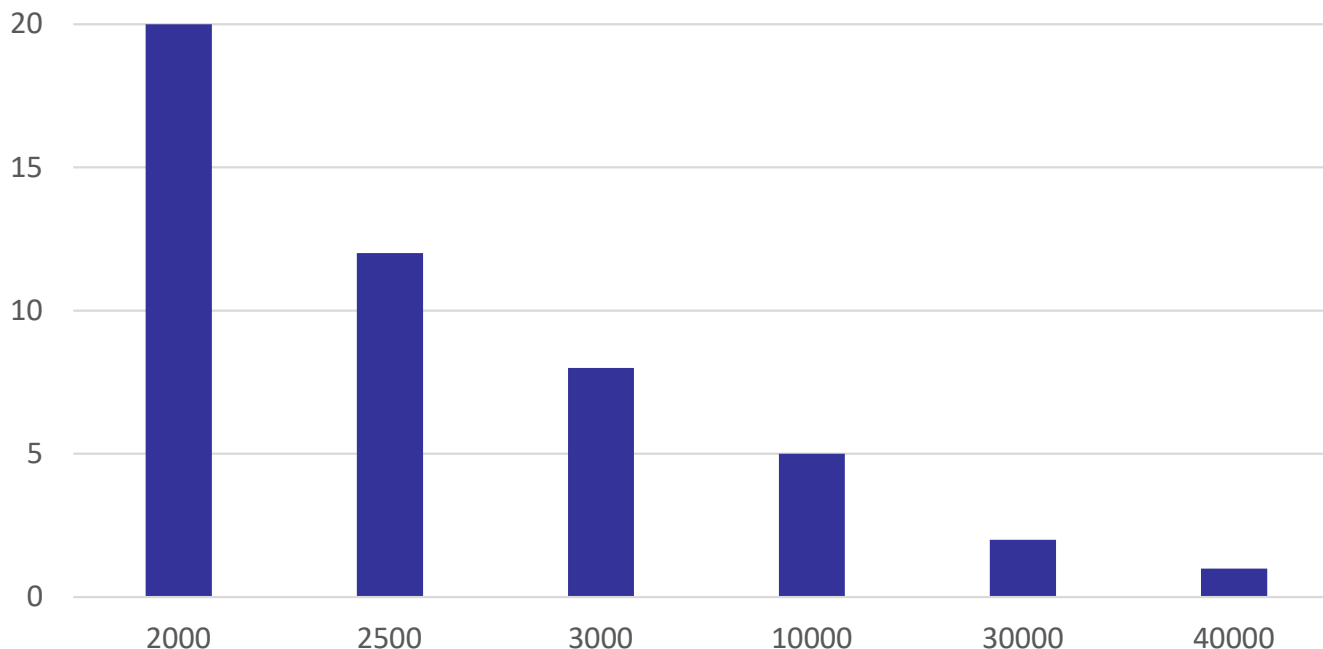
$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

- Median divides the data into two equal parts, so that the number of observations below and above the median are equal.
- Mode is the value that is observed most frequently.

A shoe manufacturing business

- 3 partners: CEO, production manager, sales manager
- 5 engineers
- 40 workers
- Average salary in this business?

Frequency of salaries



Mean	5083
Median	2500
Mode	2000

What kind of average?

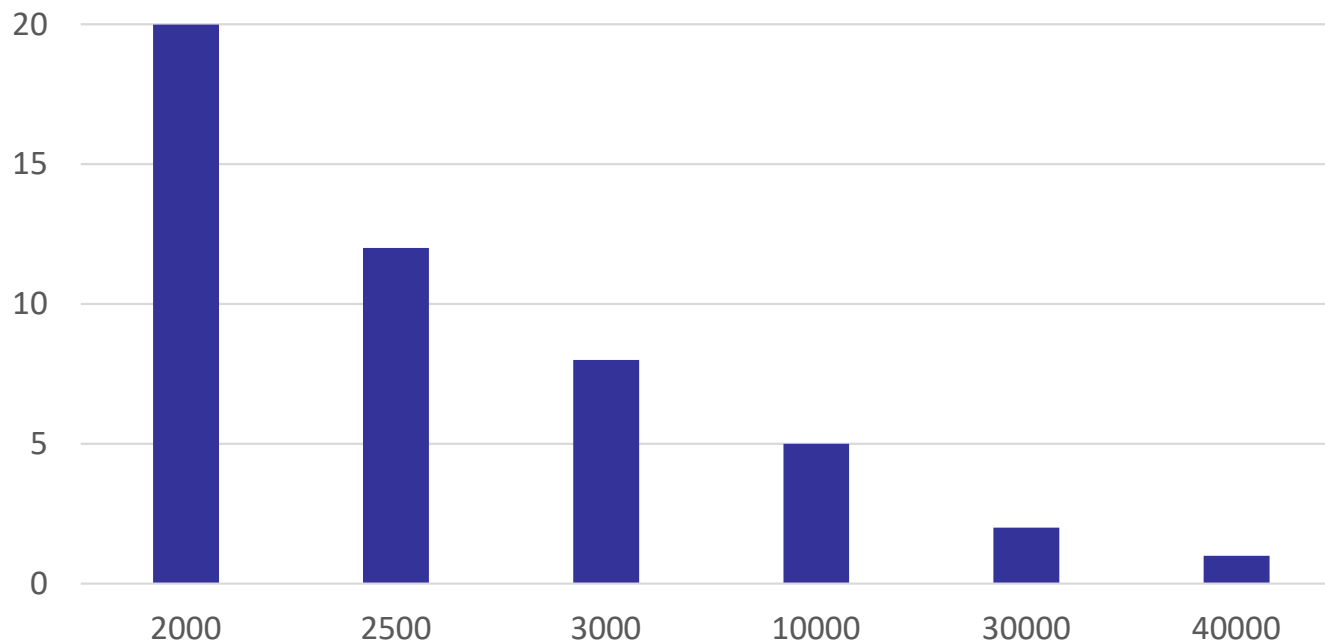
- It makes a difference!

A shoe manufacturing business

- 3 partners: CEO, production manager, sales manager
- 5 engineers
- 40 workers
- Average salary in this business?

What if 40000 is changed to 20000?
What are the new values of mean,
median and mode?

Frequency of salaries



Mean	5083
Median	2500
Mode	2000

Mean	4667
Median	2500
Mode	2000

Changes in these measures

- Mean is very sensitive to very high and very low values.
- Median is much more stable.

Sample mean

Eight prototype units for engine connectors are produced. Their pull-off forces are measured. If the pull-off force is too low, the connector may fail when it is installed in an engine.

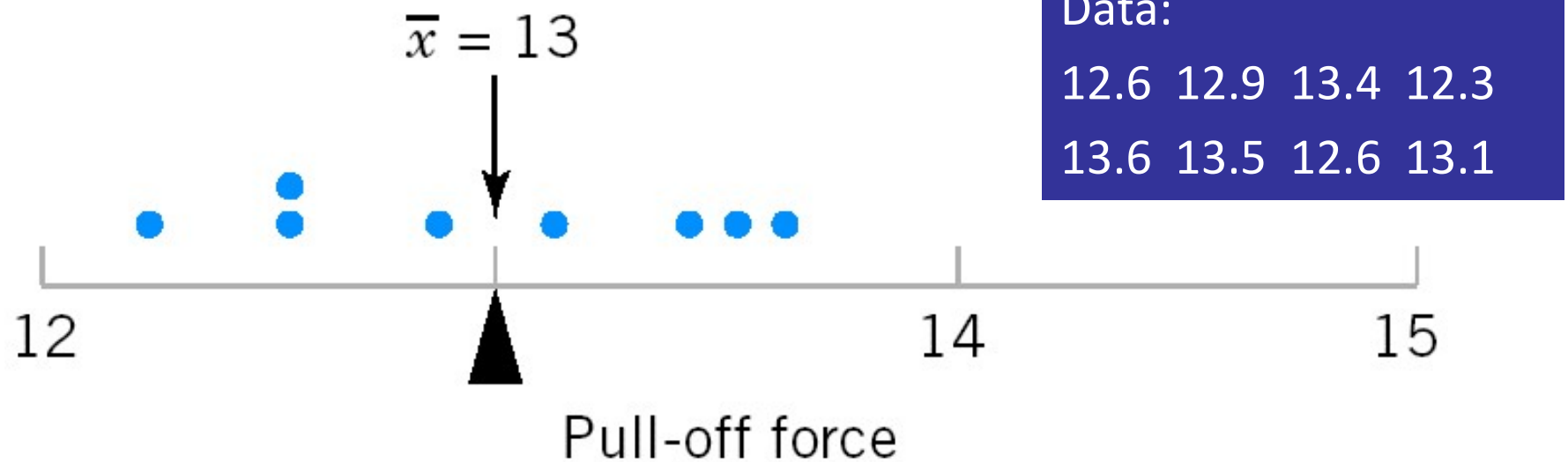
Data:

12.6 12.9 13.4 12.3

13.6 13.5 12.6 13.1

$$\begin{aligned}\bar{x} &= \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^8 x_i}{8} = \frac{12.6 + 12.9 + \cdots + 13.1}{8} \\ &= \frac{104}{8} = 13.0 \text{ pounds}\end{aligned}$$

Figure



The sample mean as a balance point for a system of weights.

Definitions:

Measures of central tendencies

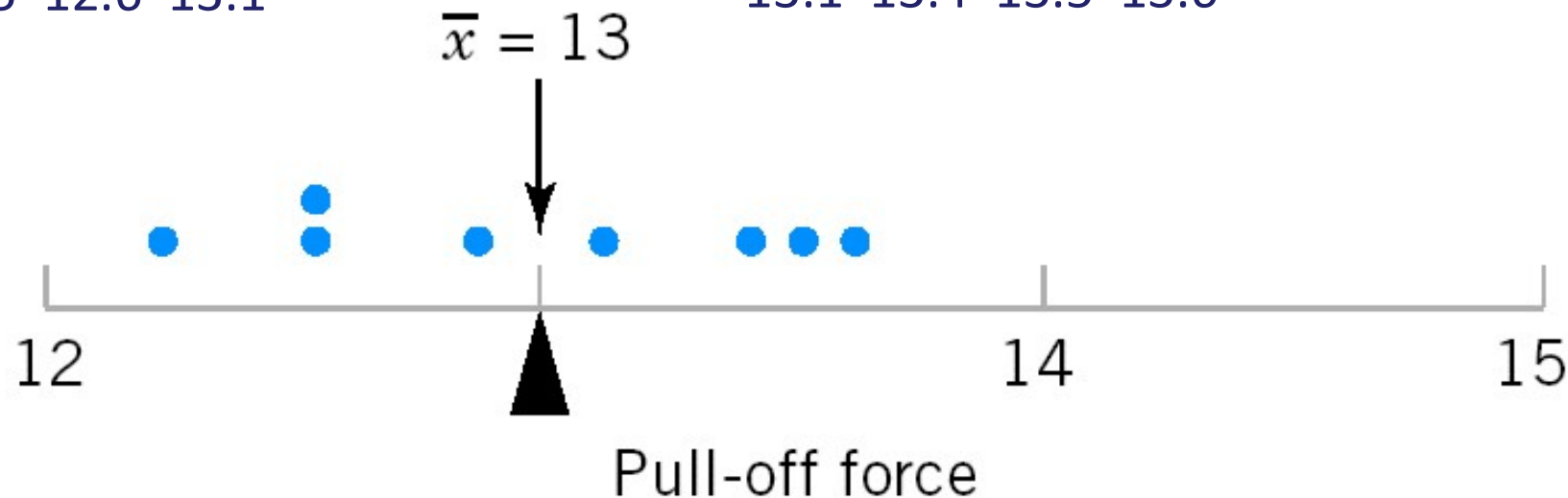
- Mode is the value that is observed most frequently.
- Median divides the data into two equal parts.

Data:

12.6 12.9 13.4 12.3
13.6 13.5 12.6 13.1

Sorted data:

12.3 12.6 12.6 12.9
13.1 13.4 13.5 13.6



Finding the Sample Median

Order the n data points from smallest to largest. Then

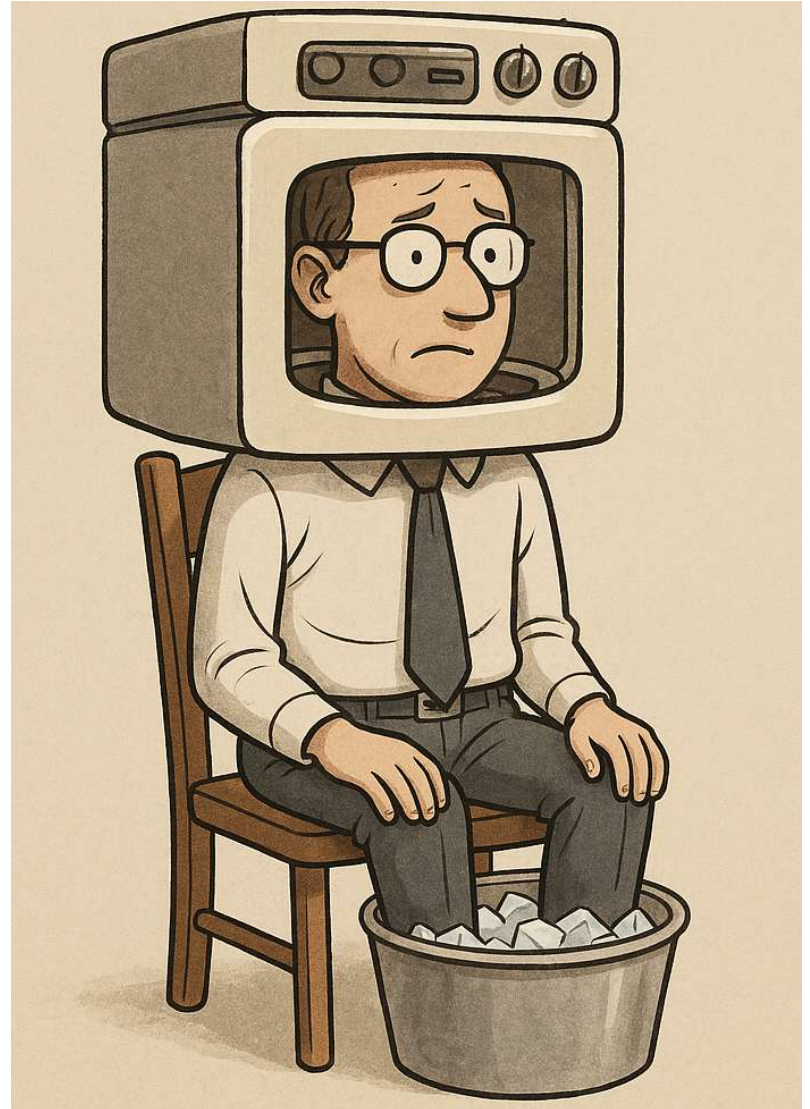
- If n is odd, the sample median is the number in position $\frac{n+1}{2}$.
- If n is even, the sample median is the average of the numbers in positions $\frac{n}{2}$ and $\frac{n}{2} + 1$.

What kind of average?

- It makes a difference in some cases, but not for all!
- When the underlying distribution is symmetric, we expect that all three measures are close to each other.
- When the underlying distribution is ***skewed***, we expect them to be very different.

Fallacy of Averages

- Did you hear about the statistician who had his head in an oven and his feet in an ice bucket?
- When asked how he felt, he replied:
"On average, I feel just fine."



What else will you look at
in the data set?

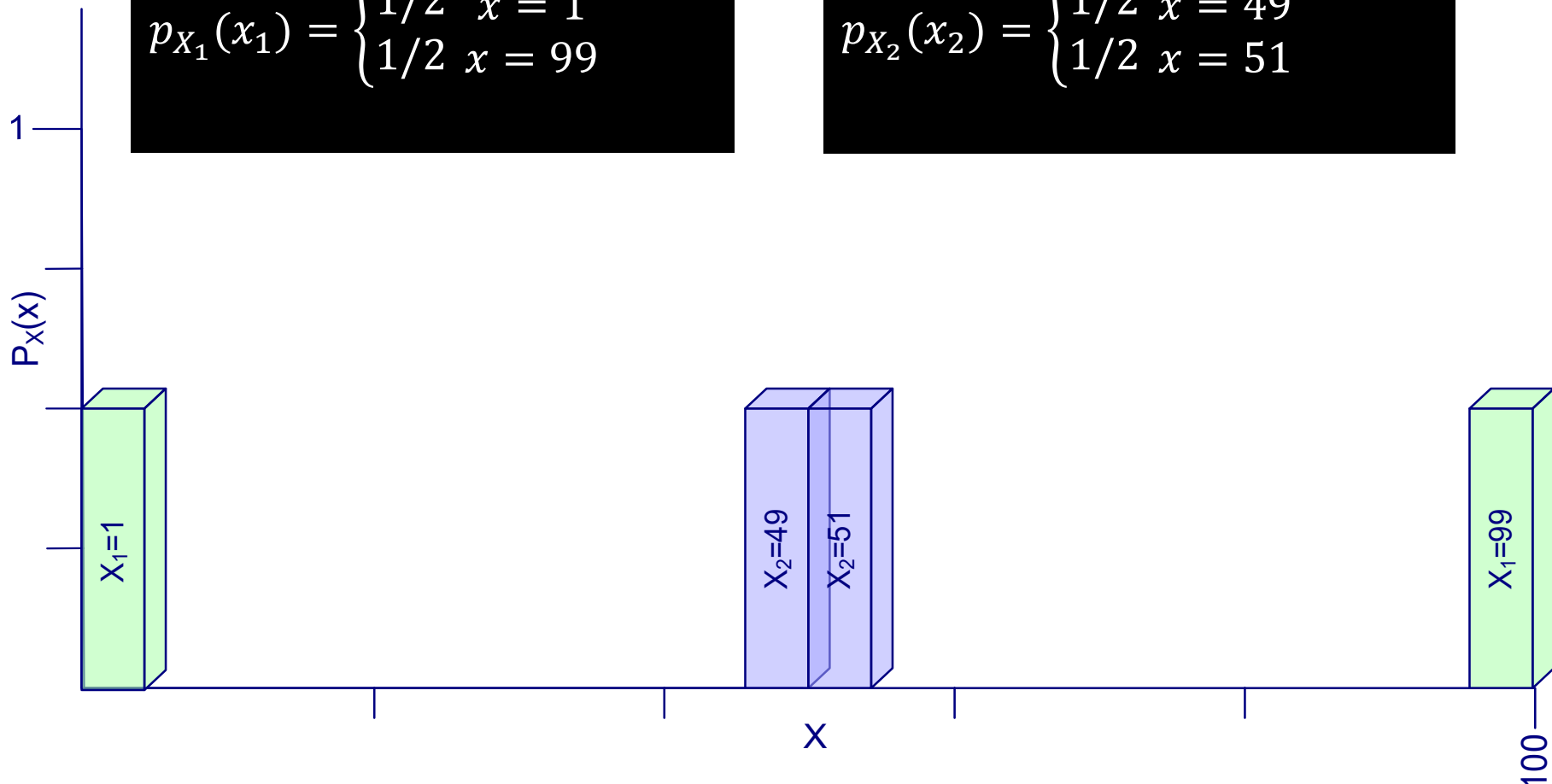
Variability?

What is variability?

- Compare the two random variables X_1 and X_2 :

$$p_{X_1}(x_1) = \begin{cases} 1/2 & x = 1 \\ 1/2 & x = 99 \end{cases}$$

$$p_{X_2}(x_2) = \begin{cases} 1/2 & x = 49 \\ 1/2 & x = 51 \end{cases}$$



Population variance

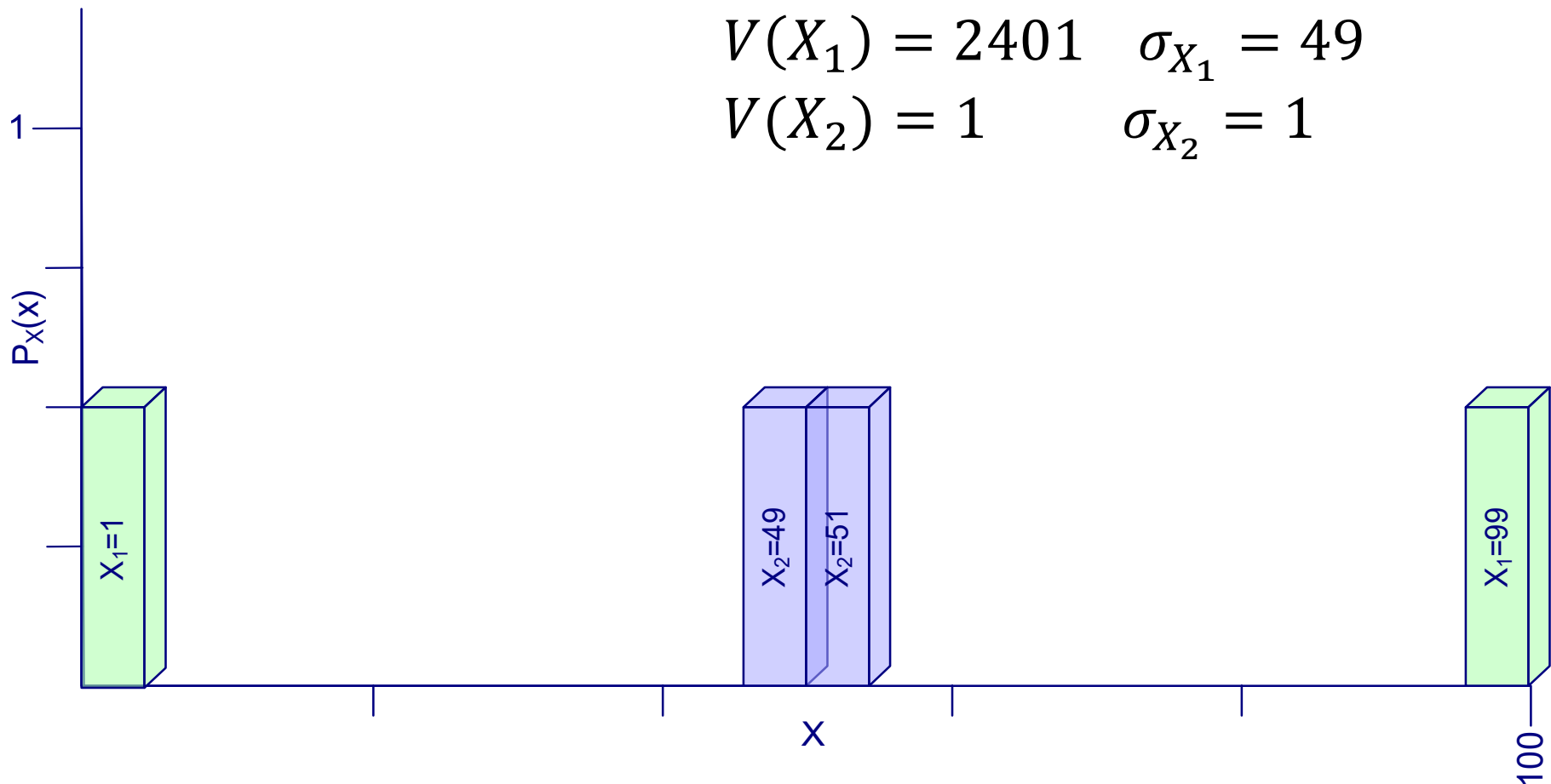
$$V(X) = E((X - \mu)^2) = \begin{cases} \sum_{\forall x} (x - \mu)^2 p(x) & X: discrete \\ \int_x (x - \mu)^2 f(x) dx & X: continuous \end{cases}$$

What does variance measure?

- Variance measures how spread the random variables are.

$$V(X_1) = 2401 \quad \sigma_{X_1} = 49$$

$$V(X_2) = 1 \quad \sigma_{X_2} = 1$$



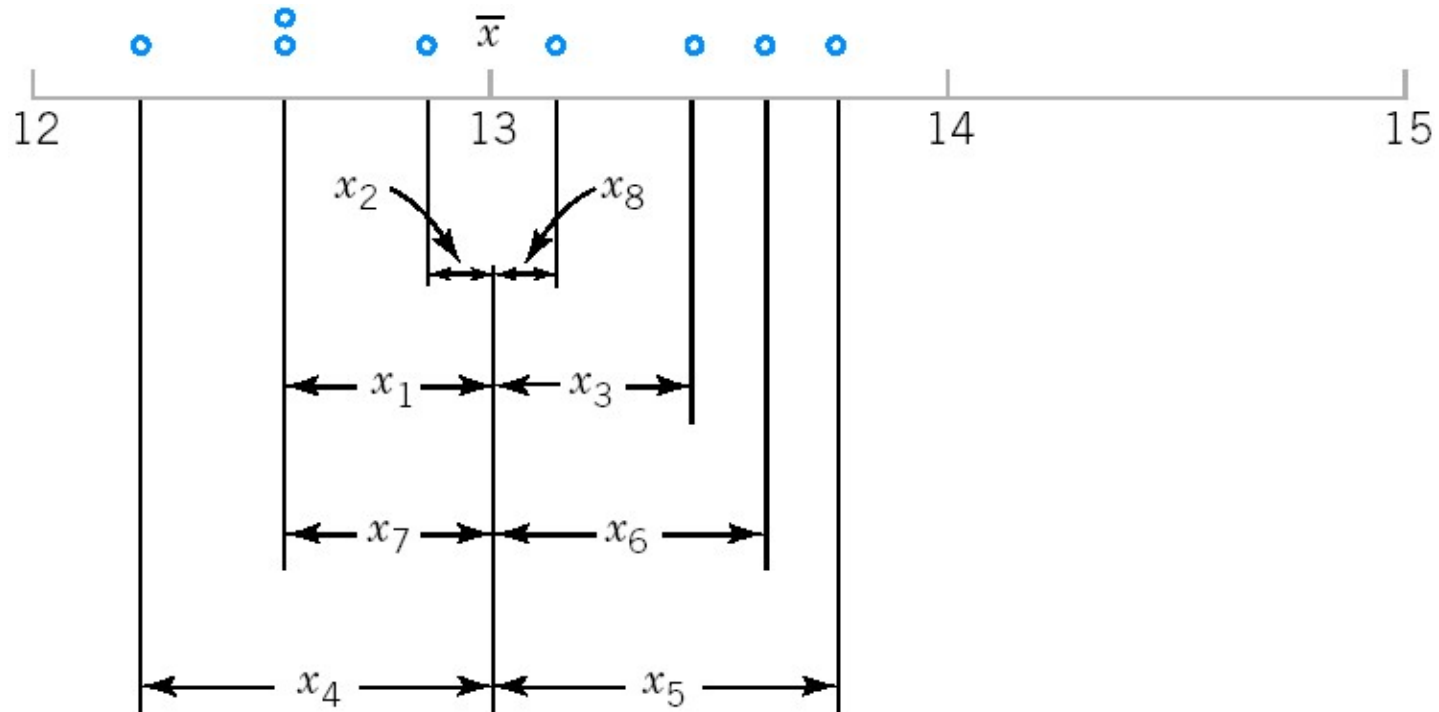
Sample Variance and Standard Deviation

If $x_1, x_2, \dots, x_n$ is a sample of n observations, the **sample variance** is

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

The **sample standard deviation**, s , is the positive square root of the sample variance.

How Does the Sample Variance Measure Variability?



How the sample variance measures variability through the deviations $x_i - \bar{x}$.

Computation of s^2

$$s^2 = \frac{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}}{n - 1}$$

$$V(X) = E((X - \mu)^2) = E(X^2) - E(X)^2$$

Sample Range

If the n observations in a sample are denoted by $x_1, x_2, \dots, x_n$, the **sample range** is

$$r = \max(x_i) - \min(x_i)$$